# Objectives

- ✓ Datatypes and Variables in Java.
- ✓ Type Conversions.

# Java is Strongly Typed

- ✓ Java is a strongly typed language since it is mandatory to have variable declared with a data type.
- ✓ This enforces certain level of restriction in a program in any operation – with the range of value a variable can hold.

# Variables in Java

- ✓ A named piece of memory where data can be stored.
- ✓ A variable is associated with a data type which determines the size of the variable.

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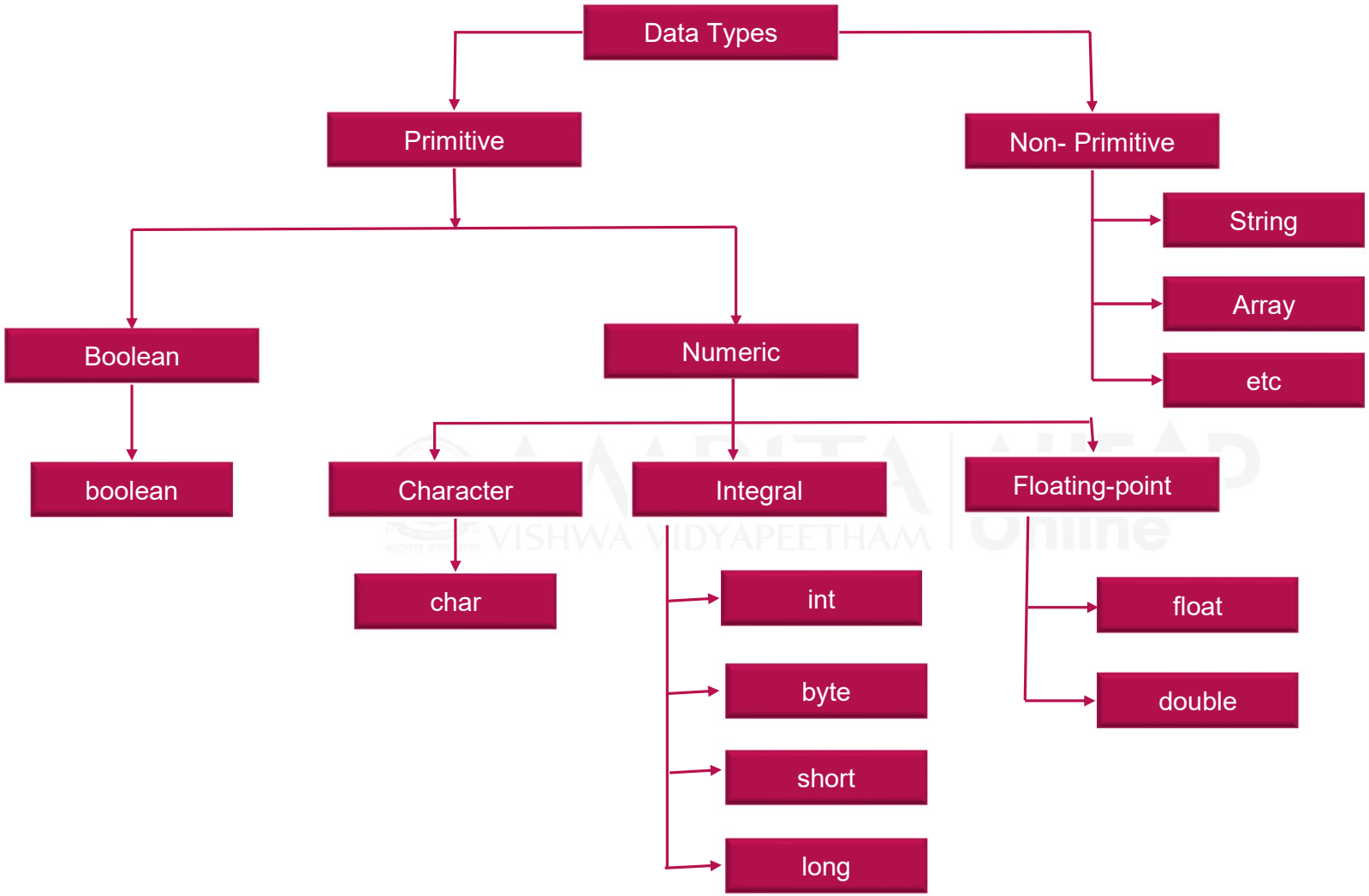
# Data Types

- ✓ Data types are important to determine the size and values that can be stored in a variable.
- ✓ Two types of data types
  - ✓ **Primitive Data Types** – building blocks of data handling.
  - ✓ **Non- Primitive Data Types** – which includes classes, interfaces and arrays.


Data Types
Primitive
Non- Primitive
String
Array
etc
Boolean
Numeric
boolean
Character
Integral
Floating-point
char
int
byte
short
long
float
double

# Primitive Data Type

- ✓ Primitive types represent single values not complex objects.
- ✓ There are eight primitive types of data:
  - ✓ byte, short, int, long, char, float, double, and boolean.
- ✓ All data types having strictly defined range that helps portability of java.
  - ✓ e.g., int is of 32 bit regardless of the platform.

# Integer

| Type | Size | Range |
|---|---|---|
| Byte | 8 bits | -128 to 127 |
| Short | 16 bits | -32,768 to 32,767 |
| Int | 32 bits | –2,147,483,648 to 2,147,483,647 |
| Long | 64 bits | -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 |

# Floating Point Types

- ✓ Real numbers with the fractional parts.
- ✓ Two types of floating-point
  - ✓ **float** - represent single precision numbers – 32 bits storage
  - ✓ **double** - represent double precision numbers - 64 bits storage

# char and boolean

**char**

- ✓ Datatype to store  a single 16-bit Unicode character.
- ✓ Range lies between '\u0000' (or 0) to '\uffff' (or 65,535 inclusive).
- ✓ Stores a single character.

**boolean**

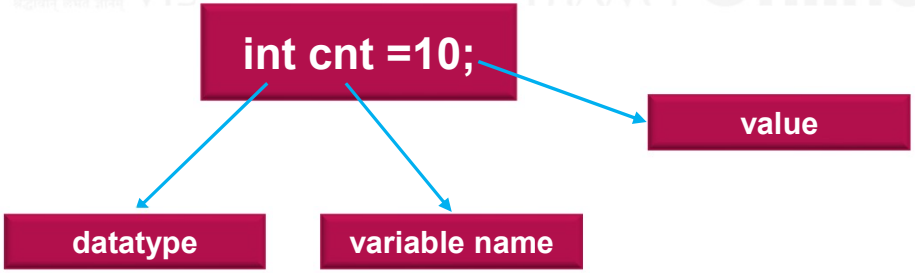
- ✓ Used for storing logical values.
- ✓ Only two possible values (True/False)

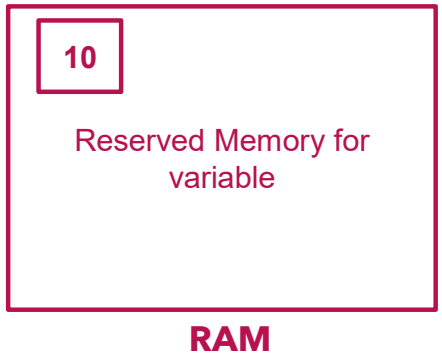
# Variables

## Declaring a Variable

- ✓ In Java, all variables must be declared before they can be used. The basic form of a variable declaration is shown here:

*type identifier [ = value][, identifier [= value] ...] ;*



# Variables

- ✓ *int a, b, c;* // declares three ints, a, b, and c.
- ✓ Java allows dynamic as well as static initializations.
- ✓ *double pi = 3.14159;* // declares an approximation of pi.
- ✓ *char x = 'x';* // the variable x has the value 'x'.

**Variables Scope**

- ✓ Scope Defined by Class.
- ✓ Scope Defined by the Methods.

# Type Conversion and Casting

Conditions for automatic conversions.

- ✓ The two types are compatible.
- ✓ The destination type is larger than the source type.

Then widening conversion takes place.

- ✓ For widening conversions, the numeric types, including integer and floating-point types, are compatible with each other.

# Casting Incompatible Types

A cast is simply an explicit type conversion.

**General form**

(target-type) value

```
int a;
byte b;
b = (byte) a;
```

- ✓ Narrowing conversions.
  e.g. Assign int variable to byte.
- ✓ Truncation
  Assigning floating point value to integer types.

# Takeaways

- ✓ Data types.
- ✓ Features of primitive data types.
- ✓ Variables.
- ✓ Type casting.